

# Paul Prae

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## PROFESSIONAL SUMMARY

AI and data engineering leader with 13+ years building production AI systems for healthcare, life sciences, and financial services. Track record of delivering enterprise solutions at AWS, Microsoft, Booz Allen Hamilton, and Slalom Consulting — from HIPAA-compliant data platforms processing petabytes of healthcare data to ML pipelines for clinical decision support covering 50M+ health plan members. Core expertise in AI agent development, cloud architecture (AWS, Azure), data engineering, and engineering team leadership. Currently building healthcare AI infrastructure and multi-agent systems at Arine, serving 45+ health plans nationwide.

## PROFESSIONAL EXPERIENCE

### Staff AI DataOps Engineer

Arine | Remote | Sep 2025 – Present

- Managed enterprise data platform on Snowflake and AWS, processing petabytes of healthcare data from hundreds of sources supporting Arine's medication-optimization platform (45+ health plans, 50M+ members)
- Developed HIPAA-compliant AI coding assistants that transformed data operations — a Data Engineering Agent that autonomously generated, deployed, and tested ETL pipelines and a Dev Environment Agent that eliminated multi-day onboarding
- Built observability pipeline for Arine's autonomous pharmacist AI agent, ingesting production behavior data from DynamoDB into Snowflake, transforming with dbt, and visualizing metrics in QuickSight
- Led AI enablement workshops training up to 100 engineers on AI-first development; designed the AI DataOps Toolkit with governance standards, reference architectures, and change management procedures

### Chief AI Architect, Senior Manager

Booz Allen Hamilton | Atlanta, GA | Jul 2024 – Mar 2025

- Led AI and analytics solution development at a Fortune 500 consulting firm (\$9B+ revenue), designing products that harnessed clinical, behavioral, and genomic data for population health and predictive medicine across FedRAMP and HIPAA-compliant environments
- Collaborated with cross-functional teams of engineers, data scientists, and domain experts to deliver AI solutions addressing algorithmic bias and data privacy risks in regulated healthcare settings
- Advised healthcare and life science clients on clinical, regulatory, and technology requirements for AI adoption in predictive medicine and research

### Neuroinformatics Data Platform Architect

TRENDS Center | Atlanta, GA | Jan 2022 – Sep 2023

- Led architecture for COINSTAC, an open-source platform for federated neuroimaging and genomics research with differential privacy, contributing to a publication in Frontiers in Neuroinformatics
- Architected AWS cloud infrastructure including GPU compute for deep learning; drove product roadmap, Agile processes, and open-source community growth through SBIR grant writing and industry partnerships
- Developed neuroinformatics tools with research scientists, translating multimodal brain imaging and omics data into biomarkers for brain health and disease

### Chief AI Officer, Founder

Hyperbloom | United States | Jan 2020 – Sep 2025

- Grew AI consulting business to \$1.4M ARR from a \$40K investment, delivering cloud-based AI and data solutions to healthcare, life science, and financial services clients
- Architected disaster recovery and data governance solutions for clinical trial software supporting 10,000+ research sites across 45 countries (5.5M remote monitoring activities monthly)
- Led teams of solutions architects and account managers through full sales cycles; trained engineers on entrepreneurship and go-to-market strategy

### Enterprise AI and ML Solutions Architect

Amazon Web Services | Atlanta, GA | Aug 2018 – May 2021

- Managed portfolio of 10+ enterprise accounts including Cox Enterprises, Cox Automotive, Equifax, NCR, Accenture, and Deloitte — delivering AI/ML architecture guidance and proof-of-concept solutions that drove production adoption
- Engaged nationally as subject matter expert in AI and machine learning; authored technical white papers, reference architectures, and presented at public and private AWS events

- Partnered with C-suite executives to define technical and business strategies including go-to-market planning, cost management, and technical recruiting

### **Senior AI Engineer**

**NeuroLex Labs** | Atlanta, GA | Jan 2018 – May 2020

- Architected automated ML pipelines for healthcare voice computing — collecting, cleaning, training, and deploying models processing text and audio data for neurodegenerative disease prediction
- Mentored international research fellows in the Tribe program; drove test-driven development adoption and authored primary technical documentation

### **Advanced Analytics Consultant**

**Slalom Consulting** | Atlanta, GA | Jul 2015 – Jan 2018

- Delivered 7+ client engagements generating \$2M+ in revenue, including a statewide behavioral health demand forecasting system across Georgia and a chatbot assessment for a Fortune 100 insurance company
- Recognized nationally as subject matter expert in NLP and machine learning; grew Slalom's AI practice through workshops, marketing, and sales support across healthcare and public sector domains

### **Software Engineer**

**Red Ventures** | Fort Mill, SC | Mar 2014 – Jul 2015

- Engineered full-stack marketing platform serving 16+ Fortune 500 clients (DirecTV, Verizon, AT&T, Liberty Mutual, Progressive); won first place at HACKATHON clt processing 140M+ rows with Apache Spark

### **Senior Support Engineer**

**Microsoft** | Charlotte, NC | Jul 2012 – Mar 2014

- Delivered advanced troubleshooting for Fortune 100 clients (Chevron, Walmart, Intel, Disney, Bank of America, BlueCross BlueShield) across hundreds of high-impact web applications; nominated for Global Business Support “Entrepreneur of the Year” (2013)

## **EDUCATION**

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### **Bachelor of Science, Computer Science (AI Emphasis) & Bachelor of**

**Arts, Cognitive Science (AI Foundation) University of Georgia** | 2008 – 2012 Graduated Cum Laude | Dean's List | Certificates in New Media and Music Business

## **TECHNICAL SKILLS**

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**AI & Machine Learning:** Generative AI, LLMs, Multi-Agent Systems, Deep Learning, NLP, Machine Learning, MLOps, Predictive Analytics, Prompt Engineering, Conversational AI

**Cloud & Infrastructure:** AWS (Bedrock, SageMaker, ECS/EKS, Lambda, Step Functions, S3, Glue, DynamoDB, QuickSight), Azure (Machine Learning, Bot Service, Cognitive Services), Snowflake, Docker, Distributed Systems

**Data Engineering:** Pipeline Architecture, CDC, ETL/ELT, dbt, Apache Kafka, PostgreSQL, Neo4j, Data Governance, HIPAA/GDPR Compliance

**Programming Languages:** Python, SQL, JavaScript, TypeScript, C#, Bash, Rust

**AI Tools & Frameworks:** Cursor AI, Claude Code, GitHub Copilot, LangChain, Ollama, n8n, Open WebUI

**Leadership & Domain:** Engineering Team Management, AI Strategy & Governance, Healthcare & Life Sciences, Technical Pre-Sales, Hiring & Mentorship, Agile/Scrum

## **CERTIFICATIONS**

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- **AWS Certified Machine Learning Engineer** — Amazon Web Services (Sep 2025)
  - **AWS Certified AI Practitioner** — Amazon Web Services (Sep 2025)
  - **AWS Certified Machine Learning – Specialty** — Amazon Web Services (Jan 2020)
  - **AWS Certified Solutions Architect** — Amazon Web Services (Jul 2018)

## **PROJECTS**

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### **Modular Earth**

Not-for-profit organization building free, open-source AI agents to help working-class entrepreneurs generate wealth. Active projects include a career strategy assistant and a financial planning agent built on a local-first, privacy-preserving stack (Ollama, LangChain, n8n, Neo4j, Tauri). Vision: deploy AI-as-a-public-service in local libraries at no cost.

## PUBLICATIONS

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[Enhancing Collaborative Neuroimaging Research: Introducing COINSTAC Vaults for Federated Analysis and Reproducibility](#)  
Frontiers in Neuroinformatics (Jun 2023). Co-authored paper presenting COINSTAC Vaults, a platform for federated neuroimaging analysis with differential privacy, enabling multi-site research collaboration without sharing raw data.